

EduVoyage Viva - 100 Questions

Use these for project viva preparation.

1. Give a 1-minute overview of your EduVoyage project architecture (frontend, backend, database).
2. What is the main problem your website solves for students?
3. Why did you choose React for the frontend instead of plain HTML/CSS/JS?
4. Why did you choose Node.js + Express for backend APIs?
5. Why did you use MySQL/MariaDB for this project?
6. Explain your project folder structure and why it is organized that way.
7. How does the login flow work from UI to database verification?
8. What data do you store in localStorage after login, and why?
9. How do JWT tokens work in your project?
10. What causes 'Invalid token' in your app and how do you fix it?
11. Why is JWT_SECRET important in backend .env?
12. What is the difference between authentication and authorization in your app?
13. How do you restrict admin/agent pages based on role?
14. What roles exist in your website now?
15. How is role assignment handled from admin panel?
16. What can an agent do that a student cannot?
17. What can an admin do that an agent cannot?
18. How does signup OTP verification work in your implementation?
19. Why should user row be created only after OTP verification?
20. How do you prevent duplicate signup for same email?
21. How do you handle password hashing and why is it necessary?
22. Which hashing algorithm/library did you use and why?
23. How does forgot-password flow work end-to-end?
24. How is reset code/token expiry managed in database?
25. What security risks exist if reset tokens never expire?
26. How did you configure Nodemailer for Gmail app password?
27. Why does Gmail reject normal password with 535 error?
28. What are common CORS issues you faced and how did you resolve them?
29. What does 'Cannot GET /api/auth/login' mean technically?
30. Why do you get 404 on /api/profile/personal sometimes?
31. What does 'Missing token' response indicate?
32. How do you test API endpoints manually (Postman/browser/Network tab)?
33. How do you debug failed requests using browser DevTools Network panel?
34. How do you read request payload and response body in DevTools?
35. Why can CSP warnings appear in console even if app works?
36. Which console errors can be ignored and which are critical?
37. How did you design the complete profile multi-step UI?
38. What are the 5 profile steps and what data each step stores?
39. How is personal information saved to database?
40. How is academic background saved to database?
41. How is study preference saved to database?
42. How are uploaded documents stored and linked to user?
43. How do you serve uploaded files from backend static path?
44. Why did document preview fail initially and how did you fix it?

45. How do foreign keys help in user_profiles/user_academics tables?
46. Why use ON DELETE CASCADE in your schema?
47. What happens if users table row is deleted?
48. How do you ensure one profile row per user (PK/unique strategy)?
49. How do you handle empty profile fields in API responses?
50. How does review/submit page combine data from all steps?
51. What should happen after profile submit and why?
52. How does your universities list page fetch paginated data?
53. Why did you keep fallback universities list in frontend?
54. What are the drawbacks of frontend fallback data?
55. How does search currently work in universities page?
56. How did you fix search so it finds results across all pages?
57. How does country filter API work for countries page?
58. Why did you create separate Countries page instead of only search?
59. How does university detail page load dynamic content by id?
60. What sections are shown on university detail page?
61. How do you handle missing university image_url?
62. How did you seed top universities data and why?
63. How do you preserve seeded universities when admin adds new ones?
64. How does admin add/edit/delete university work technically?
65. How did you improve add university form UI for better usability?
66. Why add dropdown suggestions/templates in admin form?
67. How do you avoid accidental deletion in admin panel?
68. How do you validate university form fields before submit?
69. What APIs power the scholarship finder page?
70. How do you calculate scholarship 'fit score'?
71. What student data is used for scholarship matching?
72. How could you improve scholarship recommendations in future?
73. How did you redesign scholarship UI to look more human-made?
74. How does expense tracker use selected university data?
75. How do you estimate tuition from university fee text?
76. How are suggested expenses generated by country?
77. How are expense records persisted per university?
78. What is the benefit of query param in /expense-tracker?
79. How did you connect 'Track Expenses' from university detail page?
80. What is the difference between component state and persisted storage?
81. Where do you use useEffect and why?
82. Where do you use useMemo and why?
83. How does React Router handle dynamic routes in your project?
84. What route guards/protected route approach did you implement?
85. How do you manage global auth state across pages?
86. How do you handle loading states in UI?
87. How do you handle error states in UI?
88. Give examples of friendly validation messages you added.
89. What backend logging helped you most during debugging?
90. How did you debug ETIMEDOUT database errors?
91. What caused Access denied DB errors and how did you troubleshoot?
92. What happened with Aria checksum errors in XAMPP and impact?

93. How do you recover from MySQL service crash in XAMPP safely?
94. What SQL privileges are required for your app user?
95. Why should production never use root database user?
96. What environment variables are required for your backend?
97. How do you keep secrets safe in development and deployment?
98. What are your top 3 performance improvements for this project?
99. What are your top 3 security improvements for this project?
100. If you had one more week, what major feature would you add and why?